Do not write outside the box

0 1 Potassium is in Group 1 of the periodic table.

0 1 . 1 What name is given to Group 1 elements?

[1 mark]

Tick (✓) **one** box.

Alkali metals ☐

Halogens ☐

Noble gases ☐

0 1 . 2 A potassium atom is very small.

What is the radius of a potassium atom?

[1 mark]

Tick (✓) **one** box.

0.2 cm ☐

0.2 mm ☐

0.2 nm ☐

0 1 . 3 What name is given to the number of protons in an atom?

[1 mark]

Tick (✓) **one** box.

Atomic number ☐

Group number ☐

Mass number ☐

Do not write outside the box

0 1 . 4 **Table 1** shows the name and the relative charge of the particles in a potassium atom.

Table 1

Name of particle	**Relative charge**
Proton	
	0
Electron	–1

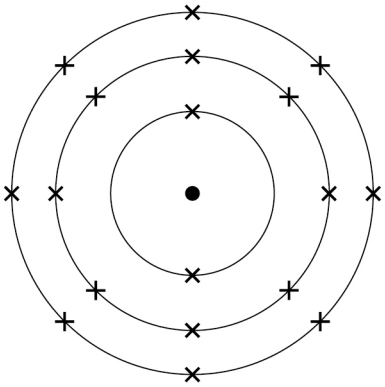
Complete **Table 1**.

[2 marks]

0 1 . 5 The electronic structure of a potassium atom is 2,8,8,1

Figure 1 represents the electronic structure of a potassium **ion**.

Figure 1



What is the charge on a potassium **ion**?

[1 mark]

Tick (✓) **one** box.

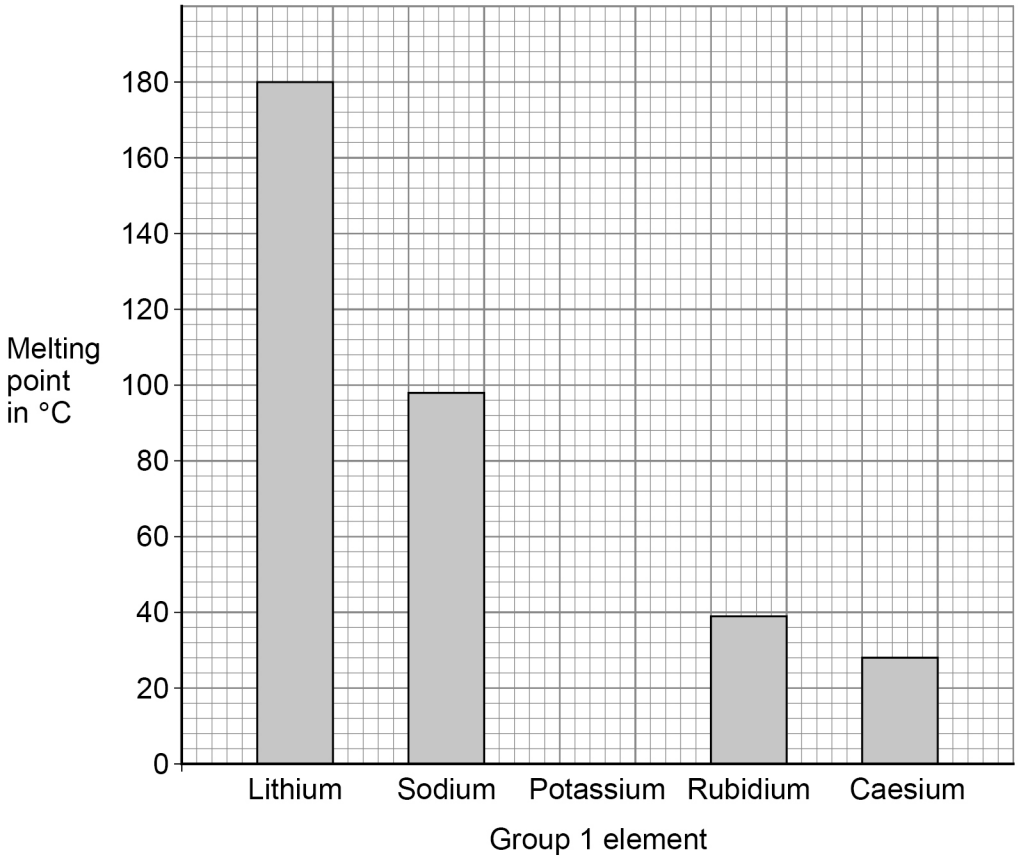
–1 ☐ 0 ☐ +1 ☐

Turn over ►

Do not write outside the box

0 1 . 6 **Figure 2** shows the melting points of some Group 1 elements.

Figure 2



Predict the melting point of potassium.

[1 mark]

Melting point = _____ °C

Do not write outside the box

0 1 . 7 Potassium has two isotopes.

Table 2 shows the percentage (%) abundance of the two isotopes.

Table 2

Isotope	Percentage (%) abundance
$^{39}_{19}K$	93
$^{41}_{19}K$	7

How is the relative atomic mass (A_r) of potassium calculated?

[1 mark]

Tick (✓) **one** box.

$A_r = \frac{(39 \times 7) + (41 \times 93)}{100}$ ☐

$A_r = \frac{(39 \times 93) + (41 \times 7)}{100}$ ☐

$A_r = \frac{(39 \times 41) + (93 \times 7)}{100}$ ☐

8

Turn over for the next question